

# WAIFINDERS WORLD

## Completion Status Report

Generated: 2026-08-09 | Platform Build: Phase 5 Production Sprint | Status: PILOT-READY

0.8818

ROC-AUC WARN Wildfire  
(Alberta)

7.15x

Top-10 Lift Open Alberta  
OGL-A Data

209

Tests Passing Disaster  
Module

7

WARN Hazard Domains  
Active in WISE

## 1. PLATFORM ARCHITECTURE

WAIFINDERS is a multi-hazard decision-support platform built on four integrated engines. All components are complete at the internal build level. Production deployment requires external data agreements, peer review, and partner engagement (see Section 7).

| Engine              | Function                                                                                    | Status                  |
|---------------------|---------------------------------------------------------------------------------------------|-------------------------|
| WARN                | External hazard scoring — wildfire, earthquake, tsunامي, flood surge, hurricane, etc.       | PILOT-READY (7 domains) |
| PULSE               | Infrastructure stress scoring — bridges, rail, water mains. Compound multi-hazard analysis. | PILOT-READY             |
| WISE                | Multi-hazard decision synthesis. Compound escalation logic. 4-tier operational state.       | PILOT-READY             |
| SENTINEL / S.A.F.E. | Federal-grade audit and evidence layer. BLAKE3-hashed outputs. Tamper-evident.              | PILOT-READY             |

## 2. WARN — HAZARD SCORING DOMAINS

WARN produces a 0–1 composite score and 4-tier operational state (NORMAL\_OPERATION / MONITOR / MITIGATION\_REQUIRED / EMERGENCY\_RESPONSE) for each hazard domain. All 7 domains are active in WISE.

| Domain             | Formula                                                                                                                          | Primary Data Source               | Anchor Result                     |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-----------------------------------|
| Wildfire           | $0.40 \times \text{FWI} + 0.25 \times \text{spread} + 0.20 \times \text{pop} + 0.15 \times \text{Cdn} + 0.00 \times \text{USFS}$ | CMIS / Canadian Forest Service    | Fort McMurray 2016: EMERGENCY     |
| Earthquake         | $0.45 \times \text{mag\_norm} + 0.35 \times \text{shallow\_norm} + 0.20 \times \text{depth}$                                     | USGS ShakeMap / ISC-GEM           | Cascadia M9.0: 0.748 MITIGATION   |
| Tsunami            | $0.45 \times \text{wave\_norm} + 0.35 \times \text{mag\_norm} + 0.20 \times \text{AOD}$                                          | NOAA DART buoy network            | 2011 Tohoku: 0.970 EMERGENCY      |
| Flood Surge        | $0.60 \times \text{surge\_norm} + 0.40 \times \text{precip\_norm}$                                                               | USGS NWIS / NOAA CO-OPS gauges    | Sandy 2.74 m: MITIGATION          |
| Hurricane          | $0.45 \times \text{wind\_norm} + 0.35 \times \text{surge\_norm} + 0.20 \times \text{precip}$                                     | NOAA HURDAT2 (1851-2023)          | Andrew 1992: 0.814 EMERGENCY      |
| Nuclear            | $0.45 \times \text{capacity\_norm} + 0.35 \times \text{epz\_pop}$                                                                | NRC Power Reactor Status (daily)  | Indian Point: 0.871 EMERGENCY     |
| Wildfire (Alberta) | Replay-tested; 14-feature PULSE form                                                                                             | Open Alberta Separately 2006-2025 | ROC-AUC 0.8818, Top-10 Lift 7.15x |

## 3. WISE — MULTI-HAZARD DECISION ENGINE

| Rule                   | Logic                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------------|
| Base tier              | max(tier_int) across all 7 active hazard domains                                                        |
| Compound escalation    | +1 tier if 2+ domains at MONITOR or above                                                               |
| Cap                    | min(EMERGENCY_RESPONSE, base + compound_boost)                                                          |
| Current session result | 6 hazards elevated → EMERGENCY_RESPONSE (compound confirmed)                                            |
| Active hazard signals  | Wildfire: 0.000 NORMAL   Earthquake: 0.399 MONITOR   Tsunami: 0.970 EMERGENCY   Flood: 0.549 MITIGATION |

#### 4. PULSE — INFRASTRUCTURE STRESS SCORING

PULSE cross-references real infrastructure datasets against authoritative hazard zones to produce a compound risk score per asset. Each domain adds a validated multiplier.

| Component                | Detail                                                                                                                                          |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Base score formula       | $0.65 \times \text{poor\_condition\_risk} + 0.35 \times \text{age\_norm}$ (per bridge)                                                          |
| Compound multiplier      | $\text{base} \times (1 + 0.25 \times \text{seismic\_HIGH} + 0.20 \times \text{tsunami\_zone} + 0.15 \times \text{flood\_SFHA})$ , capped at 1.0 |
| Seismic zones            | USGS NSHM 2014 (E2014R1) — 2%/50yr PGA at county centroid, Vs30=760 m/s. NYC metro HIGH (0.17–0.18g PG                                          |
| Tsunami zones            | NOAA CO-OPS active tide gauge presence + coastal geography (Atlantic / Sound / Inland)                                                          |
| Flood zones              | FEMA NFHL county-level SFHA (Zone A/AE) — 11 of 13 NY counties confirmed                                                                        |
| NY bridge results        | 2,000 bridges scored. RED=200 (10.0%), AMBER=340, YELLOW=1141, GREEN=319                                                                        |
| Validation (PULSE water) | Toronto AUC 0.7384 (n=4,248)   Calgary AUC 0.6626 (n=4,748) — temporal holdout                                                                  |

#### 5. WILDFIRE DOMAIN — ALBERTA REPLAY VALIDATION

This is the primary validated WARN domain. Replay executed against real Open Alberta Historical Wildfire Data (OGL-A). The synthetic ROC-AUC 0.998 (label leakage) is superseded and must not be cited.

| Parameter               | Value                        | Notes                                                  |
|-------------------------|------------------------------|--------------------------------------------------------|
| Dataset                 | Open Alberta OGL-A 2006-2025 | Dataset ID: a221e7a0-4f46-4be7-9c5a-e29de9a3447e       |
| Training events         | 20,848                       | Years 2006-2019                                        |
| Test events (held-out)  | 6,980                        | Years 2020-2025 (temporal holdout)                     |
| High-impact test events | 158                          | Large / extreme fires                                  |
| ROC-AUC (test)          | 0.8818                       | REAL DATA — replaces superseded 0.998                  |
| PR-AUC (test)           | 0.2837                       |                                                        |
| Top-10 Lift             | 7.15x                        |                                                        |
| Top-20 Lift             | 4.08x                        |                                                        |
| Mean lead time          | 274.42 hours                 |                                                        |
| Median lead time        | 17.75 hours                  |                                                        |
| Geographic scope        | Alberta only                 | Must NOT generalise to Ontario without separate replay |
| Peer review             | NOT COMPLETE                 | GATE-002 open — external dependency                    |

SUPERSEDED (do not cite):

| Metric  | Superseded Value               | Reason for Supersession                                        |
|---------|--------------------------------|----------------------------------------------------------------|
| ROC-AUC | 0.998                          | Label leakage; synthetic construction — not real wildfire data |
| PR-AUC  | 0.756                          | Same dataset; same flaw                                        |
| Dataset | Synthetic Alberta (500 events) | Not real government fire records                               |

6. GATE CLOSURE STATUS

9 production gates define the path from internal build to enterprise deployment. 2 gates are closed. 3 are blocked by external dependencies (government, third-party reviewers).

| Gate     | Name                           | Status                                   | Next Action                              |
|----------|--------------------------------|------------------------------------------|------------------------------------------|
| GATE-001 | Official Alberta Wildfire Data | PUBLIC_REPLAY_COMPLETE<br>REVIEW_PENDING | External peer review (GATE-002)          |
| GATE-002 | External Peer Review           | NOT_CLOSED<br>EXTERNAL_DEPENDENCY        | Recruit ≥3 reviewers across 6 categories |
| GATE-003 | Production Claim Gate          | PILOT_READY<br>PRODUCTION_BLOCKED        | Close GATE-001 and GATE-002              |
| GATE-004 | Water Validation Provenance    | NOT_CLOSED<br>EXTERNAL_DEPENDENCY        | Calgary, Toronto, Kitchener portals      |
| GATE-005 | Bow River Digital Twin         | NOT_CLOSED<br>EXTERNAL_DEPENDENCY        | Obtain GIS layers; Treaty 7 engagement   |
| GATE-006 | Federal SENTINEL Engagement    | READY_TO_SEND                            | Send to NRCan and Public Safety Canada   |
| GATE-007 | Deployment Operations          | CLOSED                                   | Review with pilot partner                |
| GATE-008 | External Doc Scrub             | CLOSED                                   | Re-run after any document change         |
| GATE-009 | Tests and BLAKE3 Manifest      | CLOSED (monitor for stale flags)         | Re-run after any code or output change   |

7. REMAINING GAPS — PATH TO PRODUCTION

| Priority | Gap                                 | Blocks                         | Action Required                               |
|----------|-------------------------------------|--------------------------------|-----------------------------------------------|
| CRITICAL | Official Alberta wildland fire data | Production WARN validation     | Data agreement with Government of Alberta     |
| CRITICAL | External peer review (WARN)         | 'Fully validated' claim        | Engage third-party validation organisation    |
| CRITICAL | Federal SENTINEL partner engagement | Production SENTINEL deployment | Send NRCan / Public Safety package            |
| HIGH     | Calgary water-main validation data  | Water component                | City of Calgary data agreement                |
| HIGH     | Toronto validation data             | Water component                | City of Toronto open data                     |
| HIGH     | Kitchener validation data           | Water component                | Region of Waterloo open data                  |
| MEDIUM   | Bow River Digital Twin model        | Digital Twins component        | Engineering + Bow River data feeds + Treaty 7 |
| MEDIUM   | RELATE pilot architecture           | RELATE pilot                   | Partner + architecture definition             |
| MEDIUM   | WISE knowledge corpus               | WISE pilot                     | Domain experts + data curation                |

| Priority | Gap                         | Blocks                            | Action Required                                   |
|----------|-----------------------------|-----------------------------------|---------------------------------------------------|
| LOW      | Ontario wildfire validation | Ontario WARN scope generalization | Separate Ontario replay after Alberta peer review |

## 8. APPROVED CLAIMS AND CLAIM BOUNDARY

**Approved claims (require listed qualifiers):**

- ✓ WARN is a replay-tested early-warning and risk-prioritization engine. [Qualifier: Alberta-only scope; external peer review pending; decision-support only]
- ✓ WARN has been replay-tested against real Open Alberta OGL-A public wildfire data (2006-2025). [Qualifier: peer review still pending]
- ✓ WAIFINDERS SENTINEL is available for pilot and partner discussion.
- ✓ WAIFINDERS supports Nation-controlled decision-making.
- ✓ WAIFINDERS uses BLAKE3 for all evidence hashing — no SHA, no MD5.
- ✓ PULSE provides operational stress indicators for human decision-support. [Qualifier: internal validation only]
- ✓ SENTINEL provides federal-grade audit infrastructure.

**Not approved (must not appear in any external document):**

- ✗ WARN is a fully validated forecasting system.
- ✗ WARN predicts wildfires [or any hazard outcome].
- ✗ WARN performance transfers from Alberta to Ontario.
- ✗ WAIFINDERS determines community or Nation priorities.
- ✗ AI validates Indigenous knowledge.
- ✗ PULSE autonomously directs resource dispatch.
- ✗ SENTINEL has been adopted by the federal government.
- ✗ Calgary / Toronto / Kitchener endorse WAIFINDERS.
- ✗ WARN has been independently reviewed.
- ✗ [Any citation of] ROC-AUC 0.998 — superseded, label leakage, do not cite.

## 9. TEST AND EVIDENCE SUMMARY

| Evidence Item                   | Status                      | Detail                                                                          |
|---------------------------------|-----------------------------|---------------------------------------------------------------------------------|
| WAIFINDERS Disasters test suite | 209 / 209 PASS              | 6 test files: nuclear, hurricane, earthquake, tsunami, flood, PULSE, WISE       |
| WAIFINDERS World test suite     | 73 / 73 PASS (1 stale flag) | 16 files; 1 failing test is stale-flag conflict, not fabrication (see GATE-009) |
| Alberta wildfire replay outputs | ON FILE                     | ROC-AUC 0.8818, PR-AUC 0.2837, Top-10 Lift 7.15x — BLAKE3-hashed                |
| BLAKE3 world manifest           | CURRENT                     | 175 files hashed; GATE-009 CLOSED                                               |
| External doc scrub              | PASS (0 failures)           | 101 documents scanned; GATE-008 CLOSED                                          |
| Production claim gate           | 5 / 11 CONDITIONS PASS      | PILOT_READY level; production claims blocked on external dependencies           |
| Nuclear WARN output             | 23,202 bytes                | warn_nuclear_plants.json; BLAKE3: 20e748a2b77257a6...                           |
| Hurricane WARN output           | 36,439 bytes                | warn_hurricane_events.json; 125 Cat-3+ storms, HURDAT2 1980-2023                |
| WISE decision output            | EMERGENCY_RESPONSE          | 6 of 7 hazard domains elevated; compound escalation confirmed                   |
| PULSE bridge output             | 2,000 bridges               | RED=200, AMBER=340, YELLOW=1141, GREEN=319; USGS / NOAA CO-OPS / F              |

## 10. WAIFINDERS WORLD — COMPLETED INTERNAL BUILD

All items below were completed internally. Production-deployment items require external partners.

| Component                                                                  | Status        |
|----------------------------------------------------------------------------|---------------|
| Master directory structure (00–18 + 19_EXTERNAL_READINESS)                 | COMPLETE      |
| WAIFINDERS Foundational Doctrine v2.0                                      | COMPLETE      |
| Governance Framework                                                       | COMPLETE      |
| Indigenous Data Sovereignty framework                                      | COMPLETE      |
| SENTINEL Audit & Evidence Layer                                            | COMPLETE      |
| S.A.F.E. framework                                                         | COMPLETE      |
| WARN replay validation engine (wildfire)                                   | COMPLETE      |
| PULSE wildfire stress validation engine                                    | COMPLETE      |
| Wildfire Ultra bridge                                                      | COMPLETE      |
| Full 19-step validation runner                                             | COMPLETE      |
| All validation output files (replay metrics, event CSVs, BLAKE3 manifests) | COMPLETE      |
| Claims Register (World + WARN wildfire)                                    | COMPLETE      |
| Readiness Matrix (MD + CSV)                                                | COMPLETE      |
| Source Register (CSV)                                                      | COMPLETE      |
| File Inventory                                                             | COMPLETE      |
| Dashboard shell (Streamlit, 15 tabs)                                       | COMPLETE      |
| All 16 test files (73 tests)                                               | COMPLETE      |
| Deployment Operations docs (16 documents)                                  | COMPLETE      |
| Alberta wildfire data agreement package                                    | READY TO SEND |
| External peer review package                                               | READY TO SEND |
| Federal SENTINEL engagement package                                        | READY TO SEND |
| BLAKE3 world manifest (175 files)                                          | CURRENT       |

## 11. PRIORITY NEXT ACTIONS

### IMMEDIATE — can start today

- Send Federal SENTINEL package to NRCan (Canadian Forest Service) and Public Safety Canada. Package is ready. Log outcome in FEDERAL\_CONTACT\_LOG.csv.
- Send Alberta wildfire data request letter. Use OFFICIAL\_ALBERTA\_WILDFIRE\_DATA\_REQUEST\_LETTER.md. Identify current contact at Alberta Wildfire Management Branch.
- Recruit external peer reviewers — 6 categories: Wildfire Domain (x2), ML Validation, Data Governance, Indigenous Data Sovereignty, Public Safety. Use REVIEWER\_OUTREACH\_EMAIL.md.
- Review water open-data portals: open.calgary.ca, open.toronto.ca, open.kitchener.ca. Confirm licence and permitted use. Update WATER\_PROVENANCE\_TRACKER.csv.

## NEAR TERM — requires external response

- Alberta data agreement — awaiting government response.
- Official WARN replay on Alberta wildland fire database — after agreement signed.
- External peer review — 6-12 weeks after reviewer recruitment.

## LONG TERM — multi-stakeholder

- Treaty 7 governance engagement — Nation-to-Nation process; timeline set by Nations.
- Bow River GIS layer acquisition (17 layers, multiple government sources).
- Ontario wildfire validation — separate from Alberta; separate replay required.

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CLAIM BOUNDARY — This document is an internal status report. WAIFINDERS is PILOT-READY; production and enterprise claims require external peer review and data agreements. Wildfire validation: Open Alberta OGL-A data (2006-2025); Alberta scope only; peer review pending. No results from this document may be cited as production-validated without completing GATE-002 (external peer review). BLAKE3 is the sole permitted hashing standard — no SHA-256. All WARN and PULSE outputs are decision-support tools; human operators retain all authority. Not validated for emergency management use without further review.

Generated: 2026-08-09T07:33:34 UTC | WAIFINDERS Production Engineering | BLAKE3-hashed on write